GRISHMA BALGI

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Mechanical & Electromechanical Design

EDUCATION

New York University | May 2027
M.S. Mechatronics and Robotics | GPA: 3.834/4.0

Veermata Jijabai Technological Institute | June 2025
B.Tech. Mechanical Engineering; Honors in Automobile Engineering | GPA: 9.04/10

TECHNICAL SKILLS

Design: SolidWorks, Creo Parametric, CATIA, AutoCAD, Fusion 360

Analysis & Prototyping: Ansys, mechanism design, CAD assemblies, FDM fabrication

Integration: Actuators, tendon mechanisms, fixtures, sensors, Arduino

EXPERIENCE

Prema Lab, New York University | Sep. 2025 - Present
Graduate Research Assistant | Robotics fabrication, instrumentation and simulation

- Fabricated a robotic-fin test rig and automated Silhouette Portrait 4 cutting through Python USB commands, generating coordinate-based paths for candidate fin geometries.
- Developed a motor-control interface and Arduino acquisition firmware to vary oscillation waveform, amplitude and frequency; logged load-cell readings alongside commanded and measured motor positions.
- Developing an 18-segment model of a fabricated TPU 95A tentacle in MuJoCo, representing passive bending with effective joint stiffness, damping and tendon elements.
- Iterating on geometry and passive response; investigating fluid-force effects while separating provisional model parameters from quantities requiring physical calibration.

Schneider Electric | June 2024 - June 2025
Engineering Intern | Mechanical modelling and sensor systems

- Created mechanical models and explored rigid-body dynamics and simulation-derived reduced-order models using Creo, Ansys and Altair romAI/Twin Activate.
- Used simulation data to model relationships between strain, stress and load; investigated reduced-order modelling for circuit-breaker mechanisms.

SELECTED PROJECTS

VJTI Racing | Competition Electric ATV
SolidWorks, Creo, Ansys, fabrication | Student engineering team | 2022-2024

- Contributed to mechanical subsystem design, structural analysis and fabrication for a competition electric ATV, including suspension, drivetrain mounting and chassis components.
- Supported wiring-harness and motor-controller integration, prototype fixtures, vehicle testing and fault diagnosis across iterative builds.

FlexiClimber 2.0 | Beam-Traversal Robot
C, Arduino, Parallax Propeller, SolidWorks | Course project

- Extended an existing rack-and-pinion beam-traversal platform with a pivot mechanism; integrated mechanical components, actuators and command handling for reorientation.
- Worked with Propeller C firmware for concurrent sensing, Bluetooth commands and actuator sequencing, alongside Arduino pivot-control routines; full pivot integration remained unfinished.